

Michael DeMarco

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TECHNICAL EXPERIENCE

Notion, Software Engineer

San Francisco, California—Oct 2024 to present | [Site](#)

- Overhauled the Notion database interface to boost engagement across core flows, including a **+4% in total database creations**.
- Shipped virtualization for database columns, driving a **reducing iOS webview terminations by 40%** and faster page loads.
- Boosted sub-items read performance by **50%** via novel caching scheme and introduced cycle-detection to improve product reliability.
- Contributed **10+ lint rules** to enforce high-quality and performant code organization-wide, especially with respect to AI agents.
- Rewrote early-career debugging interview used in **140+ interviews**; led onboarding sessions for new engineers.

Notion, Software Engineer Intern

San Francisco, California—May 2024 to Aug 2024 | [Site](#)

- Built developer toolbar to address client performance issues via real-time access to metrics like FPS, memory, and page load times.
- Empowered developers to tackle numerous front-end rendering issues, dramatically improving the end-user experience.
- Improved desktop tab behaviour, converted **5+ legacy class components**, and wrote ESLint rule to catch unused inline CSS.

The Canadian Space Agency, Software Engineer Intern

Saint-Hubert, Québec—May 2023 to Aug 2023 | [Site](#)

- Implemented a **genetic algorithm** for selecting waypoints based on scientific value, distance, and Lunar rover capability.
- Leveraged caching, parallelization, and data structures to reduce computation time **by 90%** for A-to-B shortest path subroutine.
- Developed multi-directional search algorithm using random trees to navigate static and dynamic obstacles between chosen waypoints.

Tesla, Software Engineer Intern

Palo Alto, California—Sep 2022 to Dec 2022 | [Site](#)

- Collaborated on internal mapping tool to assess and optimize placement of Supercharger stations around the globe.
- Led development of feature to efficiently visualize **thousands of site opportunities** with existing sites using **Flask** and **React**.
- Leveraged **React** best practices to improve code re-use in map UI and its related filters and tools, enabling future extensibility.

EDUCATION

The University of British Columbia, B.Sc. Honors Computer Science (Co-op), minor in Data Science

Vancouver, British Columbia—Sep 2019 to Apr 2024 | [Site](#)

- GPA: 93% (4.33/4.33)
- Coursework: Computer Hardware and Operating Systems, Intermediate Algorithm Design and Analysis, Internet Computing
- Awards: Kleiner Perkins Fellow, Neo Scholar Finalist, Interact Fellow, Undergraduate Teaching Assistant Award (x2), Trek Excellence Award (Top 5%) (x2), Presidential Scholar

PROJECTS

CheetCode, proof-of-concept to allow interviewees to cheat technical interviews via large language model (LLM) responses

Waterloo, Ontario—Sep 2022 to present | [Project](#) | [GitHub](#)

- Built proof-of-concept app to solve technical problems via LLM responses, undetected, using **JavaScript**, **Python**, and GPT-4.
- Earned **finalist distinction at Hack The North 2023** out of field of **250** project submissions.

VOLUNTEERING

Major League Hacking, Workshop Lead

Remote—Sep 2020 to present | [Site](#)

- Led **5 workshops** at hackathons country-wide for **400+ participants** on topics such as **React** and **machine learning**.
- Developed engaging delivery style that actively involves participants throughout workshop; recordings have garnered **500+ views**.

Cloud Native Computing Foundation (CNCF), Kubernetes Release Shadow

Remote—May 2023 to Aug 2023 | [Site](#)

- Curated and edited release notes from relevant pull requests for the v1.28 release of Kubernetes.
- Coordinated with special interest groups (SIGs) to ensure release note accuracy and determined major themes.

TECHNICAL SKILLS & INTERESTS

Languages	Python, TypeScript, JavaScript, Java, SQL, NoSQL, GraphQL, C, C++, HTML, CSS, Dart, TeX, Bash, Protobuf
Frameworks	React, Node, Flask, Django, pandas, NumPy, scikit-learn, pytest, Spring, JUnit, Tailwind
Other	Git, Docker, Kubernetes, Cloud Computing, AWS, GCP, REST, gRPC
Interests	Linguistics, Physics, Economics, Soccer, Piano, Hockey, Music, Saxophone